



CA - Experiment 4 - Practical Exercise: Build and Run a Java Application with Maven, Migrate the Same Application to Gradle

🎯 Part 1: Create and Build a Java Application with Maven

🔧 Step 1: Create a Maven Project in IntelliJ IDEA

1. **Open IntelliJ IDEA**
 - Launch **IntelliJ IDEA** and click on **File** → **New** → **Project**. ☀
2. **Select Maven**
 - In the **New Project** window, choose **Maven** from the options on the left.
 - Check **Create from archetype** and select **maven-archetype-quickstart**.
 - Click **Next**. 🧑
3. **Enter Project Details**
 - **GroupId**: `com.example`
 - **ArtifactId**: `MVNGRDLDEMO`
 - Click **Next** and then **Finish**. ✨
4. **Wait for IntelliJ to Load Dependencies**
 - IntelliJ will automatically download the Maven dependencies, so just relax for a moment. 😊

📝 Step 2: Update `pom.xml` to Add Build Plugin

To compile and package your project into a `.jar` file, you need to add the **Maven Compiler** and **Jar** plugins. 🛠

1. Open the `pom.xml` file. 📄
2. Add the following inside the `<project>` tag:

```
1 <build>
2   <plugins>
3     <!-- Compiler Plugin -->
4     <plugin>
5       <groupId>org.apache.maven.plugins</groupId>
6       <artifactId>maven-compiler-plugin</artifactId>
7       <version>3.8.1</version>
8       <configuration>
9         <source>1.8</source>
```

```
10         <target>1.8</target>
11     </configuration>
12 </plugin>
13
14 <!-- Jar Plugin -->
15 <plugin>
16     <groupId>org.apache.maven.plugins</groupId>
17     <artifactId>maven-jar-plugin</artifactId>
18     <version>3.2.0</version>
19     <configuration>
20         <archive>
21             <manifest>
22                 <mainClass>com.example.App</mainClass>
23             </manifest>
24         </archive>
25     </configuration>
26 </plugin>
27 </plugins>
28 </build>
29
```

Step 3: Build and Run the Maven Project

1. Open IntelliJ IDEA Terminal

Press **Alt + F12** to open the terminal. 

2. Compile and Package the Project

Run the following commands to build the project:

```
1 mvn clean compile
2 mvn package
3
```

3. Locate the JAR File

After running the above, your `.jar` file will be located at:

```
1 D:\Idea Projects\MVNGRDLDEMO\target\MVNGRDLDEMO-1.0-SNAPSHOT.jar
2
```

4. Run the JAR File

To run the generated JAR file, use:

```
1 java -jar target\MVNGRDLDEMO-1.0-SNAPSHOT.jar
2
```

Part 2: Migrate Maven Project to Gradle

Step 1: Initialize Gradle in Your Project

1. Open Terminal in IntelliJ IDEA

Make sure you're in the project directory:

```
1 cd "D:\Idea Projects\MVNGRDLDEMO"
```

2

2. Run Gradle Init Command

Execute the following command to migrate your Maven project to Gradle:

```
1 gradle init --type pom
2
```

This command will convert your Maven `pom.xml` into a Gradle `build.gradle` file. 🛠️

🛠️ Step 2: Review and Update `build.gradle`

1. **Open** `build.gradle` in IntelliJ IDEA.
2. Ensure the following configurations are correct:

```
1 plugins {
2     id 'java'
3 }
4
5 group = 'com.example'
6 version = '1.0-SNAPSHOT'
7
8 repositories {
9     mavenCentral()
10 }
11
12 dependencies {
13     testImplementation 'junit:junit:4.13.2'
14 }
15
16 jar {
17     manifest {
18         attributes(
19             'Main-Class': 'com.example.App'
20         )
21     }
22 }
23
```



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COMMIT TO ACHIEVE

🛠️ Step 3: Build and Run the Gradle Project

1. Clean and Build the Project

To clean and build your Gradle project, run:

```
1 gradle clean build
2
```

2. Run the Generated JAR File

Now, run the generated JAR file using:

```
1 java -jar build/libs/MVNGRDLDEMO-1.0-SNAPSHOT.jar
2
```